

Sr.No	Module	Weightage %	Marks Weightage%
1	Module-I	20.00	23.68
2	Module-V	20.00	21.05
3	Module-III	20.00	19.74
4	Module-IV	20.00	19.74
5	Module-II	20.00	15.79

Weightage representation okay ☐ : Weightage representation exceeded more than 5% ☐ :Weightage representation exceeded more than 10% ☐

CSE601

[ET]

Enroll. No. - _____

END SEMESTER EXAMINATION : April-May, 2026

Data Structures and Algorithm Design

Time : 3.00 Hrs

Maximum Marks : 60

Note: Attempt questions from all sections as directed.

Section - A : Attempt any Four questions out of Five . Each question carries 06 marks.

Q1.

(a)

Define a Binary Search Tree (BST) and its search complexity.

(3)

Delete

(b)

What is an AVL Tree? Explain the concept of "Balance Factor."

(3)

Delete

Q2.

Discuss and develop Java code for a Singly Linked List.

(6)

Delete

Q3.

(a)

Explain Circular Linked List and its operations.

(3)

Delete

(b)

Define the principle of "Optimal Substructure" in Dynamic Programming.

(3)

Delete

Q4.

(a)

Explain the Stack Data Structure with an example.

(3)

Delete

(b)

Write Java Code for push() operations of a stack represented using a linked list.

(3)

Delete

Q5.

Solve the Fractional Knapsack problem using the Greedy method.

(6)

Delete

Section – B : Attempt any two questions out of three. Each question carries 10 marks.

Q6.

(a)

Provide a step-by-step trace of Kruskal’s Algorithm on a sample graph.

(5)

Delete

(b)

Explain the single-source shortest path problem using Dijkstra’s algorithm.

(5)

Delete

Q7.

(a)

Discuss Circular Queue with an example.

(5)

Delete

(b)

List out the time complexities of all the sorting algorithms.

(5)

Delete

Q8.

(a)

Develop a Java code snippet for the insert operation in a BST.

(5)

Delete

(b)

Explain the characteristics of NP-Hard problems.

(5)

Delete

Section - C : Compulsory question

Q9.

(a)

Explain a Doubly Linked List and its operations.

(4)

Delete

(b)

Write Enqueue and Dequeue operations.

(4)

Delete

(c)

Develop Java code for the insert operation in the General Binary Tree.

(4)

Delete

(d)

Explain the Knuth-Morris-Pratt (KMP) string matching algorithm and the role of the prefix function.

(4)

Delete